

Quiz 3 Solutions: Applied Natural Language Processing DATA 441/641

1. Why might one choose to use a GRU over an LSTM?
 - A) GRUs consistently outperform LSTMs in all sequence modeling tasks.
 - B) GRUs include an additional memory gate compared to LSTMs.
 - C) GRUs are less sensitive to hyperparameter tuning than LSTMs.
 - D) GRUs require fewer parameters and are computationally more efficient.**
2. What is the purpose of max pooling in a CNN?
 - A) To selectively emphasize higher-weighted neurons in the convolutional layers.
 - B) To reduce the spatial dimensions of feature maps while retaining important information.**
 - C) To remove all non-linearity from feature maps.
 - D) To perform a learnable form of dimensionality reduction.
3. A CNN applies a 3×3 filter with stride 1 on a 32×32 grayscale image without padding, followed by a 5×5 filter with stride 2 and no padding. What will be the final output dimension?
 - A) 28×28
 - B) 16×16
 - C) 13×13**
 - D) 10×10
4. What is the primary role of the forget gate in an LSTM cell?
 - A) To determine which parts of the input should be ignored before entering the cell.
 - B) To clear the hidden state at each step for preventing memory overflow.
 - C) It determines how much past information should be erased from the cell state.**
 - D) To decide the importance of each feature map within the cell.
5. Why might an RNN or LSTM be used in image processing?
 - A) To efficiently process images as a single-dimensional sequence.
 - B) To process images with a sequential structure, such as handwritten text recognition.**
 - C) To detect edges and textures in images.
 - D) To perform data augmentation on image datasets.
6. What is the primary advantage of using convolutional layers instead of fully connected layers for image processing?
 - A) Convolutional layers ensure that each pixel contributes equally to feature learning.
 - B) They completely eliminate overfitting in deep learning models.
 - C) They reduce the number of parameters by sharing weights.**

- D) They replace dropout layers in network architectures.
7. What is the primary issue with standard RNNs when dealing with long sequences?
- A) They require additional attention layers for memory retention.
 - B) They can only process sequences up to a fixed length.
 - C) They introduce bias due to sequential updates in hidden states.
 - D) Vanishing and exploding gradients.**
8. Which of the following best describes how RNNs handle sequential data?
- A) They pass information forward without storing any past states.
 - B) They rely entirely on convolutional layers to process dependencies.
 - C) They maintain a hidden state that carries information from previous time steps.**
 - D) They encode positional embeddings similar to Transformers.
9. What is a key difference between GRUs and LSTMs?
- A) GRUs introduce an additional cell state to improve memory retention.
 - B) GRUs are a purely feedforward mechanism without recurrent components.
 - C) GRUs have fewer gates and do not use a separate cell state.**
 - D) GRUs are strictly designed for NLP tasks and cannot be used for time series.
10. In the context of CNNs, why do deeper networks often perform better?
- A) They replace standard convolutions with self-attention mechanisms.
 - B) They eliminate the need for fully connected layers.
 - C) They extract more complex hierarchical features.**
 - D) They prevent overfitting by using a larger number of filters.
11. Which of the following best describes the primary role of convolution in a CNN when applied to text data?
- A) To merge words into a fixed-length representation for fully connected layers.
 - B) To identify global semantic patterns in entire text sequences.
 - C) To extract local patterns and n-gram features.**
 - D) To sequentially process each word in the text using recurrence.
12. Which of the following is a major challenge when applying RNNs or LSTMs to images?
- A) They cannot be trained using standard backpropagation algorithms.
 - B) They do not perform well on sequential data.
 - C) They cannot model non-sequential dependencies.
 - D) RNNs are inherently designed for 1D sequential data, making them inefficient for processing high-dimensional images.**
13. In an LSTM, why do we use a cell state in addition to a hidden state?
- A) The cell state helps directly compute gradients to avoid vanishing gradients.
 - B) The cell state replaces the hidden state in standard RNNs.
 - C) The cell state ensures that all past information is stored indefinitely.
 - D) To store long-term memory independently from short-term activations.**